

Machine Learning Problem Statement

EV Range Prediction from Specifications — Regression

Challenge theme: Turn a static EV specification dataset into a reproducible machine-learning solution that predicts vehicle range from specifications and can be tested on new user inputs.

1. Challenge Overview

Participants will work with a supplied electric-vehicle specification dataset and build a complete regression solution for predicting EV range. The challenge is designed to evaluate the complete machine-learning workflow—not only the final model. Teams are expected to demonstrate sound data cleaning, exploratory analysis, feature engineering, model development, evaluation, reproducibility, documentation and a working interactive demonstration.

The dataset is a static specification sheet. It does not contain trip-level telemetry such as traffic, weather, HVAC usage, State of Charge or State of Health. Therefore, the challenge is specifically a specification-based range prediction problem rather than real-time range estimation.

2. Evaluation Structure

The hackathon will be conducted in two evaluation rounds. The first round will evaluate submitted work, while the second round will provide an opportunity for shortlisted teams to present and demonstrate their solution in physical presence.

- Round 1 — Unstop Submission Evaluation: Teams will submit the required solution materials through Unstop. The notebook, technical report, data preparation, EDA, feature engineering, model performance, pipeline, code quality, innovation and other applicable criteria will be evaluated using the rubric in this document.
- Round 2 — Physical Presentation & Technical Evaluation on EV Day (9 September): Shortlisted teams will be evaluated in physical presence on EV Day on 9 September. Teams will present their solution to the judges, explaining the problem approach, data cleaning, EDA, feature engineering, model selection, training, evaluation, pipeline and key results. Teams may also be asked to demonstrate their trained model and provide live inputs so that judges can verify the model's output and understand how the solution responds to different specifications.

Round 2 is intended to assess both the team's understanding of its submitted work and the technical quality of the solution. Teams should be able to clearly justify why each necessary step was performed, why the selected model was chosen, how it was evaluated, and how the final pipeline produces predictions. Judges may also inspect or test the model output during the presentation.

3. Dataset

The supplied dataset contains 478 electric vehicle models from 59 brands, with one row per model. It includes battery, performance, charging, dimensions, drivetrain, body type, seating, cargo and towing specifications.

Field / Group	Description
brand / model	Manufacturer and model name

battery_capacity_kWh	Usable battery capacity
battery_type	Lithium-ion for all 478 rows; zero variance and not intended as a predictive feature
number_of_cells	Battery cell count; missing in 202 of 478 rows (about 42.3%)
torque_nm	Motor torque; missing in 7 rows (about 1.5%)
efficiency_wh_per_km	Energy consumption; may be used for EDA/sanity checks, but NOT as a final model input
range_km	Official driving range; regression target
top_speed_kmh	Top speed
acceleration_0_100_s	0–100 km/h acceleration time
fast_charging_power_kw_dc	Maximum DC fast-charging power; missing in 1 row
fast_charge_port	Charging connector; almost all rows are CCS and not intended as a key predictive feature
towing_capacity_kg	Towing capacity; missing in 26 rows (about 5.4%)
cargo_volume_l	Cargo volume; includes 3 non-numeric 'Banana Boxes' entries requiring cleaning
seats	Number of seats
drivetrain	AWD, FWD or RWD
segment	15 market-segment categories
length_mm / width_mm / height_mm	Vehicle dimensions
car_body_type	8 body-style categories
source_url	Source listing metadata; not intended as a predictive feature

The target and core specification features are sufficiently populated for this problem. Known data-quality issues are deliberately part of the challenge and must be identified and addressed.

The dataset contains 478 rows, which is workable for classical machine-learning regression but relatively small for deep learning. Teams are encouraged to prioritize appropriate regularized models and tree-based ensembles such as Random Forest, Gradient Boosting, XGBoost or LightGBM, although other suitable approaches are permitted.

4. Problem Statement — EV Range Prediction from Specifications

Problem Statement

Electric-vehicle range is influenced by more than battery capacity alone. Vehicle dimensions, drivetrain, motor characteristics, performance and overall vehicle design contribute to how effectively an EV uses its stored energy. The challenge is to learn these relationships from the available specifications and predict the official driving range of an EV.

Teams must build a regression model that can take an EV's specifications as input and estimate its range without simply reconstructing the target from an efficiency formula.

Objective

Build a regression solution that predicts:

Target variable: range_km — official driving range in kilometres.

The model may use relevant specifications such as:

- Battery capacity and, where appropriately handled, number of cells
- Motor torque, top speed and 0–100 km/h acceleration
- Fast-charging power
- Drivetrain, segment and body type
- Length, width and height
- Seats, cargo volume and towing capacity
- Participant-created features derived from the available columns

Important restriction: efficiency_wh_per_km MUST NOT be used as a model input for the final range predictor. Battery capacity and efficiency are directly related to range, so including efficiency would allow the model to largely reconstruct the target algebraically. efficiency_wh_per_km may still be used during EDA and for post-prediction sanity checks.

5. Required Machine-Learning Workflow

1. Understand and profile the dataset before modelling.
2. Identify missing values, duplicates, inconsistent data types and unusual entries.
3. Clean and preprocess the data with clear justification for each decision.
4. Perform meaningful EDA using appropriate numerical and categorical analyses and visualizations.
5. Create, select and justify useful features. Feature engineering is strongly encouraged.
6. Split the data appropriately into training and testing sets and avoid target leakage.
7. Train and compare multiple suitable regression models.
8. Tune the selected model where appropriate.
9. Evaluate the final model using appropriate regression metrics such as MAE, RMSE and R^2 .
10. Build a reproducible preprocessing + model pipeline and save the trained artifact.
11. Create an interactive tester where users/judges can enter EV specifications and receive a predicted range.

6. Model Flexibility

Teams are not restricted to a single algorithm. Suitable approaches may include Linear, Ridge or Lasso Regression, Decision Trees, Random Forest, Gradient Boosting, XGBoost, LightGBM, CatBoost, Support

Vector Regression, k-NN regression, neural networks and ensemble or stacking methods. The chosen approach must be appropriate for the dataset and technically justified.

Given the dataset size, deep neural networks are not expected to be automatically advantageous. Teams should demonstrate why their chosen model is suitable rather than assuming that a more complex model is better.

7. Feature Engineering & Innovation

Teams are encouraged to go beyond directly feeding raw columns into a model. Feature engineering can be used to capture relationships that may be useful for predicting range.

- Create physically or domain-inspired features from existing specifications.
- Combine dimensions into useful size/footprint proxies where justified.
- Create interaction or ratio features where they have a meaningful interpretation.
- Handle categorical variables appropriately.
- Explore alternative feature-selection strategies and justify the final feature set.
- Use imputation strategies thoughtfully; `number_of_cells` may be included as an optional engineered feature, but its high missingness must be addressed.

Innovation must be grounded in the supplied dataset. Do not fabricate vehicle specifications or use hidden target information to artificially improve performance.

8. Evaluation Criteria

Evaluation Criterion	Weight
Data Cleaning & Preprocessing	15%
Exploratory Data Analysis	10%
Feature Engineering & Innovation	15%
Model Performance — Regression Accuracy	25%
Pipeline & Reproducibility	10%
Code Quality & Documentation	10%
Technical Report	10%
Interactive Demo / Live Testing	5%
TOTAL	100%

9. Submission Requirements

Each team must submit all of the following through the designated Unstop submission process:

Submission	Requirement
1. Solution Notebook	Jupyter Notebook or Google Colab notebook containing data loading, inspection, cleaning, EDA, feature engineering, model training, evaluation and final results.
2. Technical Report	A concise technical report explaining what was done and why: cleaning decisions, EDA findings, feature choices, model selection, evaluation results, limitations and trade-offs.
3. Trained Model	The final trained regression model must be saved as a reusable artifact, for example <code>.pkl</code> or <code>.joblib</code> .
4. Reproducible Pipeline	Preprocessing and the trained model should be packaged into a reproducible pipeline so new inputs undergo the same transformations used during training.

5. Interactive Testing Tool	A final user-facing interface must allow a user/judge to enter EV specification values and obtain a predicted range.
6. Supporting Files	Any required dataset, requirements/environment information, helper scripts, configuration files or application files needed to reproduce and test the solution.

10. Notebook & Code Quality Standards

- The notebook must be well structured and readable from top to bottom.
- Every major step should contain a short explanation of what is being done and why.
- Use descriptive variable names such as `battery_capacity`, `range_target`, `numeric_features` and `trained_model` rather than `x`, `df2`, `temp` or `a1`.
- Comments should explain non-obvious logic; comments should not simply repeat the code.
- EDA must support decisions made later in the workflow rather than consist only of decorative plots.
- Fix random seeds where appropriate so that results can be reproduced.
- Respect train/test separation and avoid target leakage.
- Model selection should be supported by metrics and reasoning, not only by choosing the highest score.

11. Interactive Final Demonstration

The final solution must include a simple interface so that a judge can test the trained solution without editing the underlying model code.

Judge enters	System returns
Allowed EV specifications such as battery capacity, torque, top speed, acceleration, dimensions, seats, drivetrain, body type and other features used by the final pipeline.	Predicted EV range in kilometres.

The interface may be implemented using a notebook widget, Gradio, Streamlit or another lightweight interface. During the physical evaluation on EV Day, 9 September, judges should be able to modify input values and see the model respond immediately.

12. Final Evaluation — EV Day

Shortlisted teams will participate in the second evaluation round in physical presence on EV Day on 9 September. This round will primarily be a presentation and technical evaluation of the submitted solution. Teams will explain why they performed their necessary data, EDA, feature engineering, modelling and evaluation steps and will present their final results.

- Present the problem, objective and overall solution approach.
- Explain the data-cleaning and preprocessing decisions and why they were necessary.
- Explain the EDA findings and how they influenced subsequent modelling decisions.
- Explain the feature engineering choices and why the final feature set was selected.
- Explain the models considered, the final model selection and the reasoning behind it.
- Present evaluation results, metrics and limitations honestly.

- Demonstrate the trained model/pipeline and, where requested, provide live input values so judges can verify the model output.
- Answer technical questions about preprocessing, features, model selection, evaluation, reproducibility and the final prediction pipeline.
- Show the data-cleaning and EDA decisions that materially affected the solution.
- Explain the feature engineering and why the final model was selected.
- Show evaluation results and discuss limitations honestly.
- Demonstrate the saved pipeline and trained model.
- Use the interactive tester live with judge-provided input specifications.
- Answer technical questions about preprocessing, features, model selection, evaluation and reproducibility.

13. Expected Outcome

The final solution should be a reproducible machine-learning system that accepts an EV's specifications and produces a defensible predicted range. Strong submissions should demonstrate a complete and technically sound workflow from raw data to a deployable prediction pipeline.

The emphasis of the challenge is not simply on achieving the best numerical score. Strong solutions should show clear reasoning, meaningful EDA, effective feature engineering, appropriate model selection, clean implementation, reproducibility, good documentation and a convincing live demonstration.

14. Challenge Flow

**RAW DATA → CLEANING → EDA → FEATURE ENGINEERING → REGRESSION →
EVALUATION → PIPELINE → LIVE TEST**

**ROUND 1: UNSTOP SUBMISSION → SHORTLIST → ROUND 2: PHYSICAL EVALUATION ON EV
DAY — 9 SEPTEMBER**